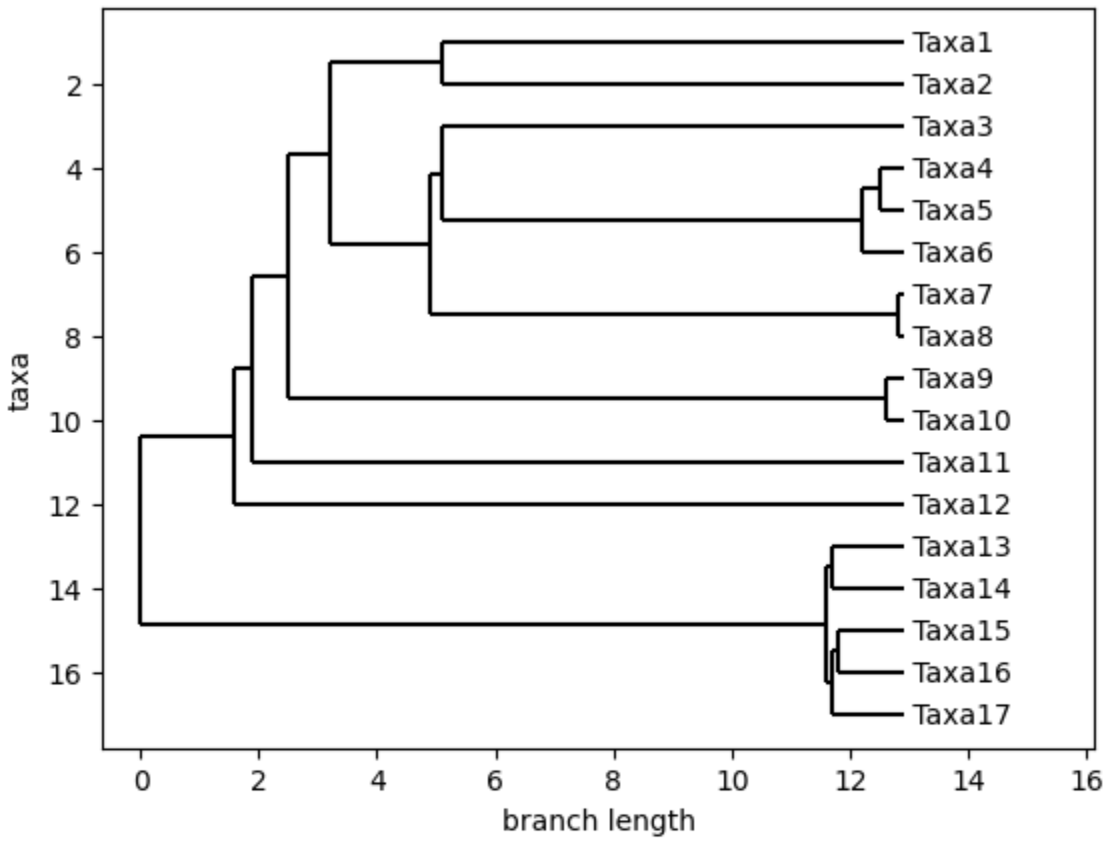
In [11]:
```
import tskit
import msprime
from Bio import Phylo
from io import StringIO
```

In [12]:
```
# species tree of all annual taxa (branch length in MYr)
spTree = "((((((Taxa1:7.8,Taxa2:7.8):1.9,((Taxa3:7.8,((Taxa4:0.4,Taxa5:0.4):
```

In [13]:
```
spTree_phylo=Phylo.read(StringIO(spTree),"newick")
Phylo.draw(spTree_phylo)
```



In [14]:
```
# msprime based on the species tree demography
# recombination_rate of 0.5 makes each tree independent (different genes)
# sequence_length then specify number of trees
demography = msprime.Demography.from_species_tree(
    spTree,
    time_units="myr",
    initial_size=10**4,
    generation_time=1)
ts = msprime.sim_ancestry({"Taxa1": 1, "Taxa2": 1,"Taxa3":1,"Taxa4":1,"Taxa5
                          "Taxa7":1,"Taxa8":1,"Taxa9":1,"Taxa10":1,"Taxa11":
                          "Taxa13":1,"Taxa14":1,"Taxa15":1,"Taxa16":1,"Taxa1
                          random_seed=123,sequence_length=100,recombination_
ts
```

Out[14]:

ts kit **Tree Sequence**

Trees	100
Sequence Length	100.0
Time Units	generations
Sample Nodes	17
Total Size	184.9 KiB
Metadata	No Metadata

Table	Rows	Size	Has Metadata
Edges	3200	100.0 KiB	
Individuals	17	500 Bytes	
Migrations	0	8 Bytes	
Mutations	0	16 Bytes	
Nodes	1617	44.2 KiB	
Populations	33	1.5 KiB	✅
Provenances	1	13.6 KiB	
Sites	0	16 Bytes	

In [15]:

```
t1=ts.first()
t1.draw_svg(y_axis=True,size=[600,600])
```

Out[15]:

In [16]:
```
#drop mutations (rate can be varied)
ts_mutated = msprime.sim_mutations(ts, rate=5e-8)
ts_mutated
```

Out[16]:

tskit Tree Sequence

Trees	100
Sequence Length	100.0
Time Units	generations
Sample Nodes	17
Total Size	205.6 KiB
Metadata	No Metadata

Table	Rows	Size	Has Metadata
Edges	3200	100.0 KiB	
Individuals	17	500 Bytes	
Migrations	0	8 Bytes	
Mutations	487	17.6 KiB	
Nodes	1617	44.2 KiB	
Populations	33	1.5 KiB	✅
Provenances	2	14.3 KiB	
Sites	99	2.4 KiB	

In [22]:
```
ts_mutated.aslist()[2].draw_svg(y_axis=True,size=[600,600])
```

Out[22]:

In [18]:
```
len(ts_mutated.genotype_matrix())
```

Out[18]:
```
99
```

In [23]:
```
# anything non-zero is considered as gene-loss
ts_mutated.genotype_matrix()[2]
```

Out[23]:
```
array([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0], dtype=int32)
```

In [20]:
```
import numpy as np
a=np.array(ts_mutated.genotype_matrix())
np.savetxt("/workdir/sh2246/p_phyloGWAS/output/tskit_mutatedGenotypeMatrix.t
```

In [ ]:

In [ ]:

In [ ]: